

FORMAL EXPERIMENTAL REPORT

# CTR-GCN Baseline V1

NTU RGB+D 120 Interaction-26  
Cross-Subject

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Joint stream + Bone stream + fixed equal-weight score-level fusion.  
Generated from the completed formal experiment artifacts without  
retraining or inference.

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**88.49%**

Joint Test Top-1

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**88.33%**

Bone Test Top-1

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**91.11%**

Fusion Test Top-1

# 1. Objective

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This experiment is the project's first complete formal CTR-GCN baseline for 26-class action recognition on the Interaction-26 subset. It evaluates two independently trained models—a joint-coordinate stream and a runtime-derived bone stream—and fixed equal-weight score-level fusion. Runtime probes and smoke tests are excluded.

## 2. Dataset and Split

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| Item                                        | Verified value |
|---------------------------------------------|----------------|
| Training samples                            | 11764          |
| Validation samples                          | 1308           |
| Official test samples                       | 11660          |
| Classes                                     | 26             |
| Skeleton joints ( <code>num_point</code> )  | 25             |
| Maximum persons ( <code>num_person</code> ) | 2              |

The official NTU120 CSub training partition was divided by the project split into training and validation subsets. The official CSub test partition remained untouched and was loaded only for final evaluation after all 65 epochs. **The test set was not used for model selection or hyperparameter tuning.**

## 3. Model

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CTR-GCN models channel-wise topology refinements for graph-based skeleton action recognition. The Joint model used joint coordinates ( `bone=False` , `vel=False` ). The separate Bone model used `bone=True` , `vel=False` ; its bone representation was computed by the feeder at runtime from joint coordinates, not loaded from a separately pre-generated bone dataset. These are two independent CTR-GCN models, not one six-channel model.

Their class-score vectors were combined without additional training:

```
S_fusion = S_joint + S_bone
```

## 4. Experimental Setup

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### Environment

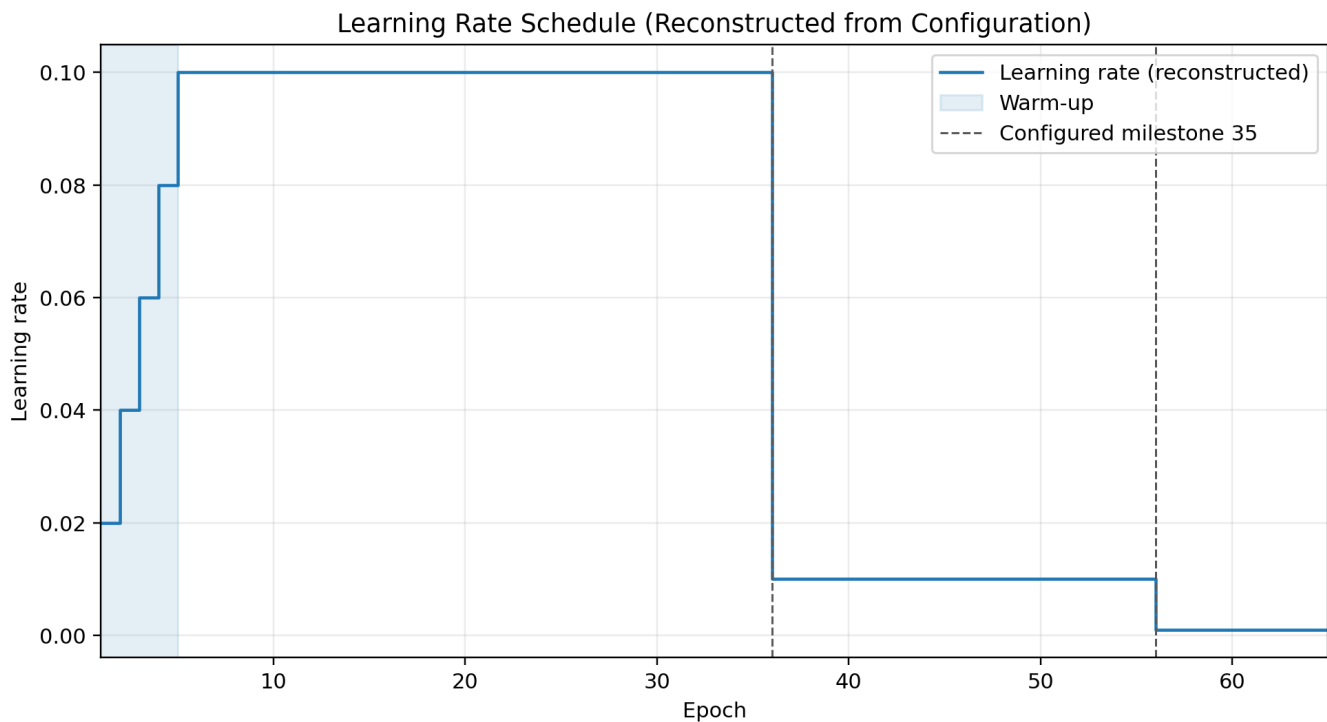
| Component                         | Value                                                             |
|-----------------------------------|-------------------------------------------------------------------|
| Python                            | 3.10.21                                                           |
| PyTorch                           | 2.6.0+cu118                                                       |
| CUDA runtime bundled with PyTorch | 11.8                                                              |
| Training hardware                 | NVIDIA GeForce RTX 3090 ×3 available; physical GPU 0 + GPU 1 used |
| Parallelism                       | PyTorch DataParallel, configured devices [0, 1]                   |

Python, PyTorch, and bundled CUDA runtime versions were verified with CPU/read-only commands. No CUDA availability probe or `nvidia-smi` command was run. Training had previously been validated and executed using two RTX 3090 GPUs.

## Training hyperparameters

| Parameter                     | Verified value                                     |
|-------------------------------|----------------------------------------------------|
| Epochs                        | 65                                                 |
| Batch size / test batch size  | 64 / 64                                            |
| Optimizer                     | SGD                                                |
| Base learning rate            | 0.1                                                |
| Weight decay                  | 0.0004                                             |
| Nesterov                      | True                                               |
| LR steps / decay rate         | [35, 55] / 0.1                                     |
| Warm-up epochs                | 5                                                  |
| Window size                   | 64                                                 |
| Seed                          | 1                                                  |
| Train augmentation            | <code>random_rot=True , p_interval=[0.5, 1]</code> |
| Validation/test preprocessing | <code>random_rot=False , p_interval=[0.95]</code>  |

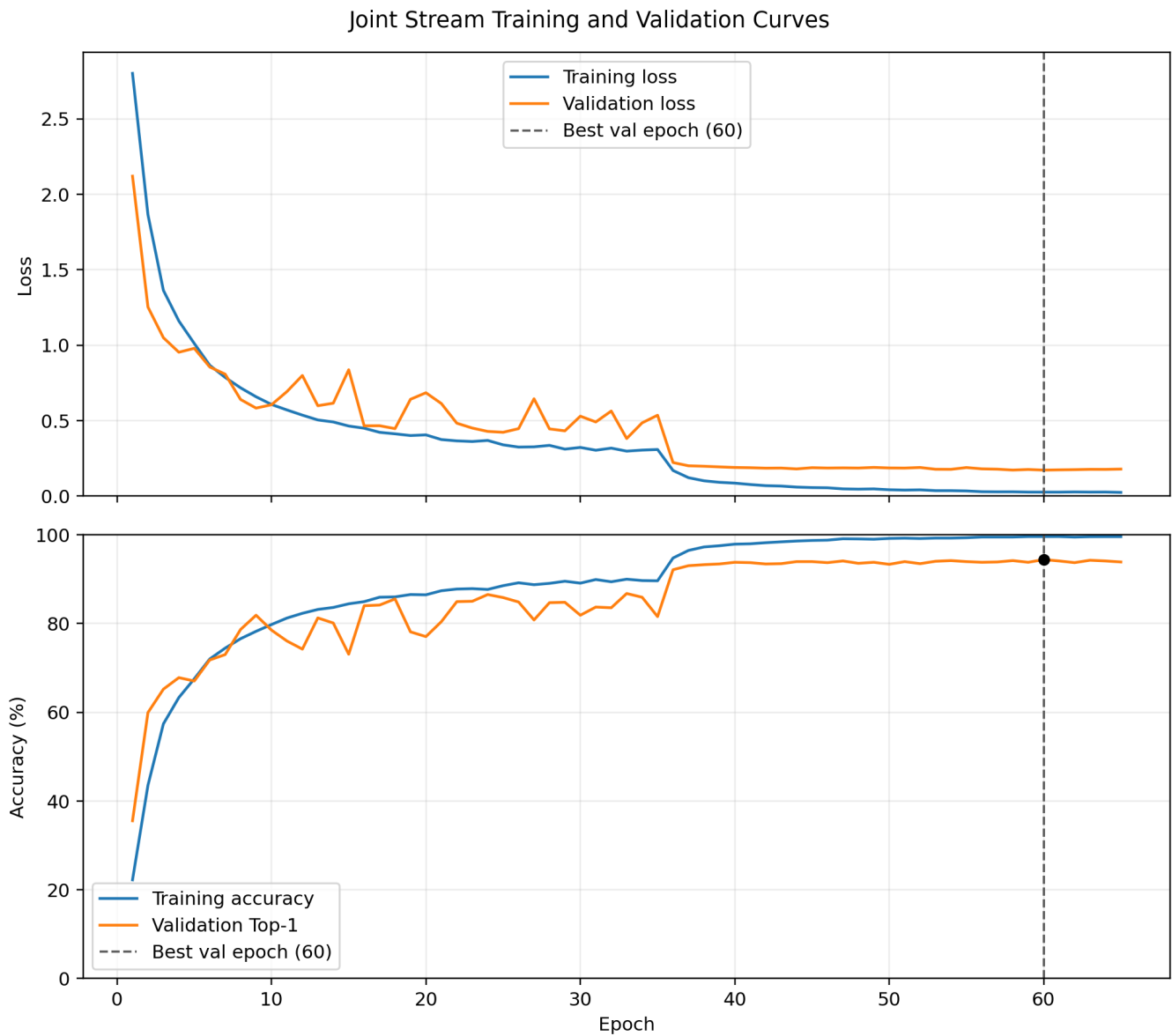
The LR series in `learning_rate_schedule.png` is **reconstructed from configuration and scheduler code**, not claimed as a measured log series; the source logs do not print LR per epoch.



## 5. Evaluation Protocol

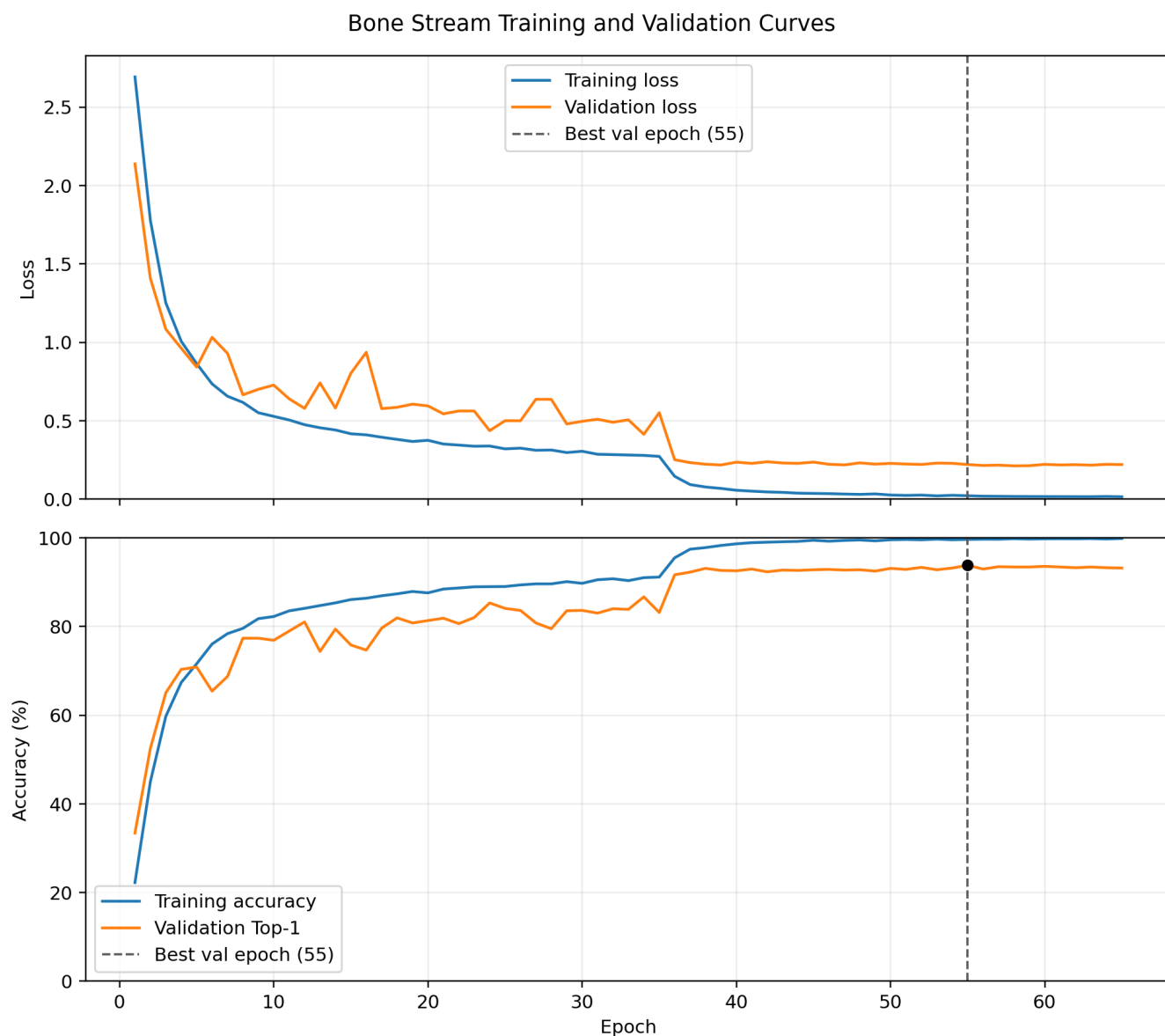
Each epoch performed training followed by validation. Validation Top-1 selected `best_model.pt`. After epoch 65, the selected checkpoint was loaded and evaluated exactly once on the official test split. Both logs contain complete epochs 1–65. The persisted score artifacts from those final evaluations were used here to independently recompute all Top-k and fusion results.

## 6. Joint Stream Results



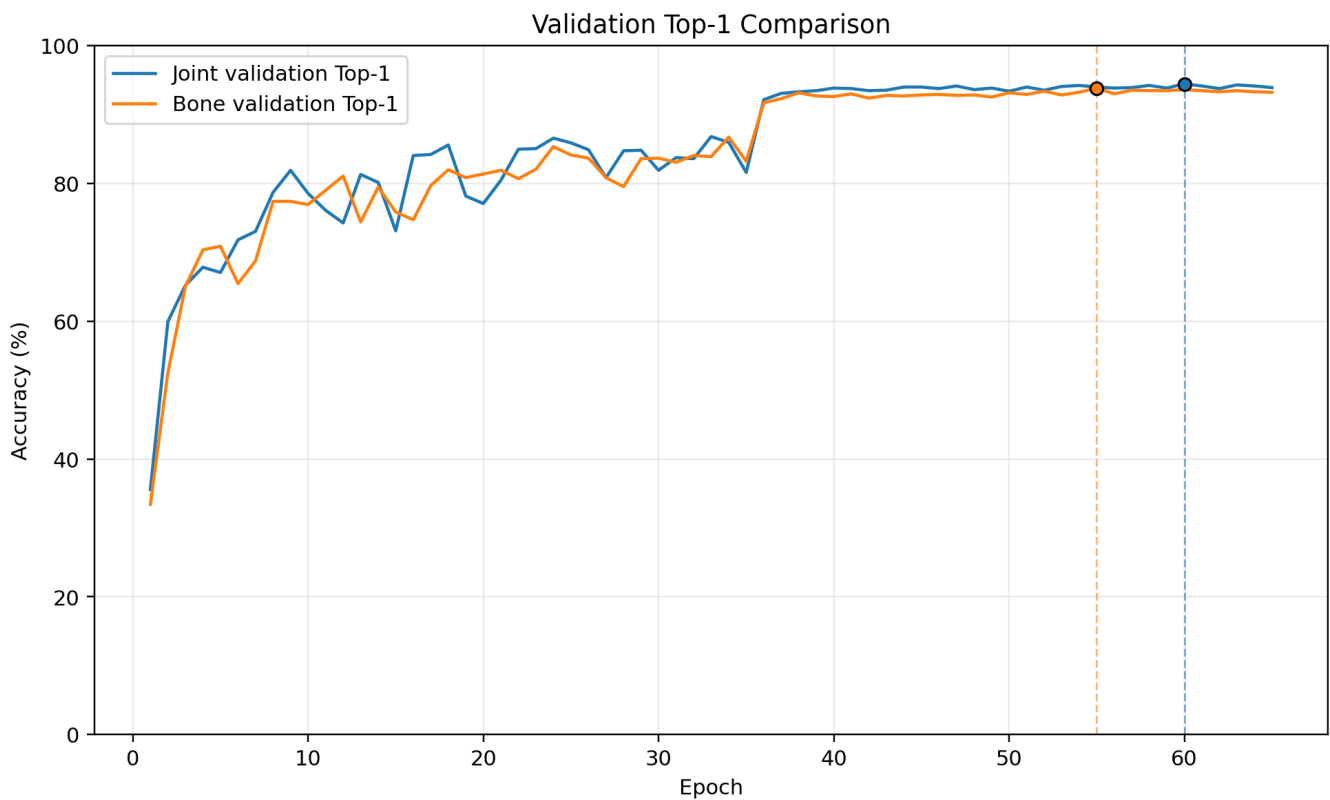
The best validation Top-1 was 94.42% at epoch 60. Official test loss was 0.400430, with Top-1 88.49% and Top-5 98.37%. Validation Top-1 ended at 93.88% after reaching its maximum at epoch 60; the late epochs remained close to the best value.

## 7. Bone Stream Results



The best validation Top-1 was 93.81% at epoch 55. Official test loss was 0.430637, with Top-1 88.33% and Top-5 98.20%. Validation Top-1 ended at 93.20% after its epoch-55 maximum.

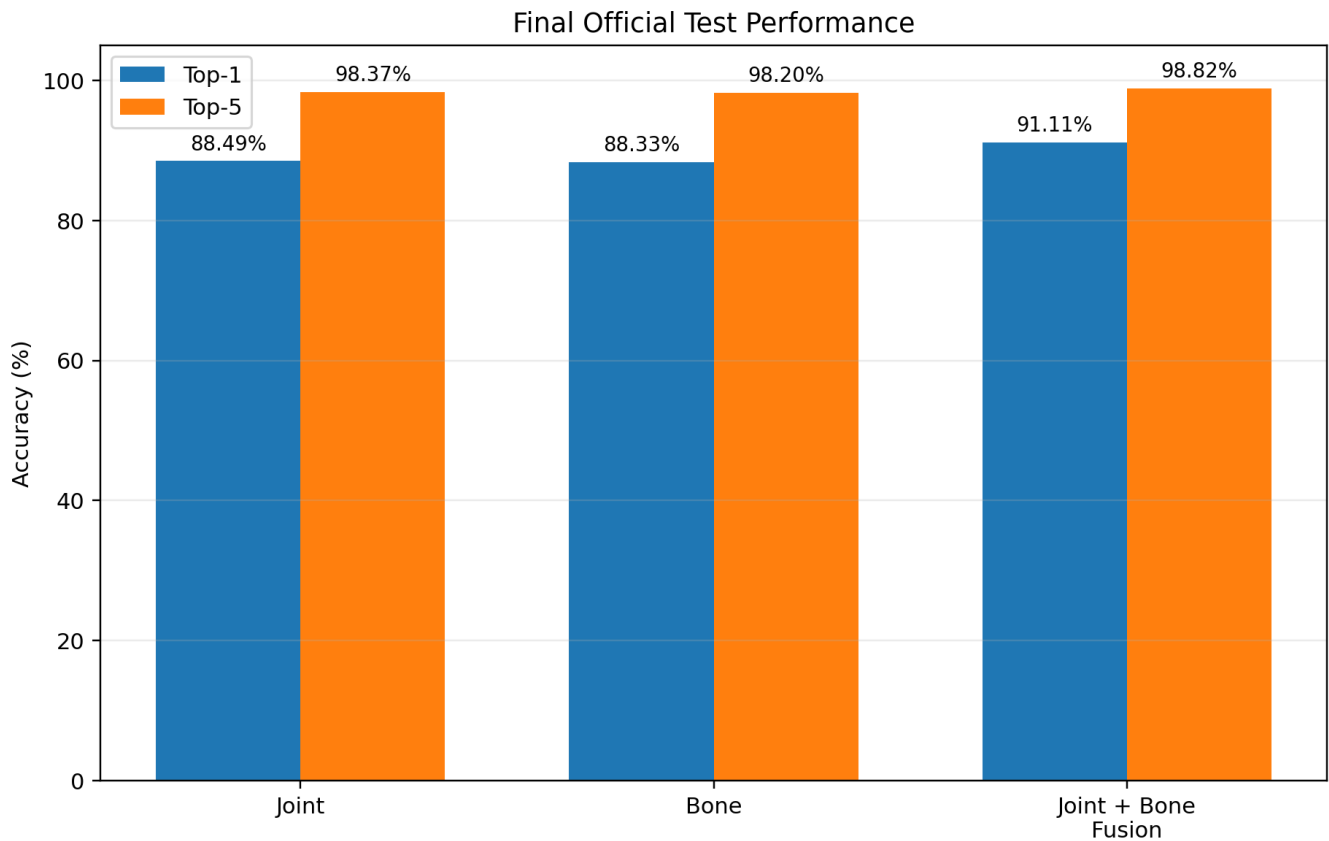
## 8. Joint vs Bone Validation Behaviour



The streams converged to similar validation performance. Bone reached its selected maximum five epochs earlier (epoch 55) than Joint (epoch 60); Joint's best validation Top-1 was 0.61 percentage points higher. The curves show a late-stage plateau with epoch-to-epoch variation; no causal interpretation is asserted.

## 9. Two-Stream Score Fusion

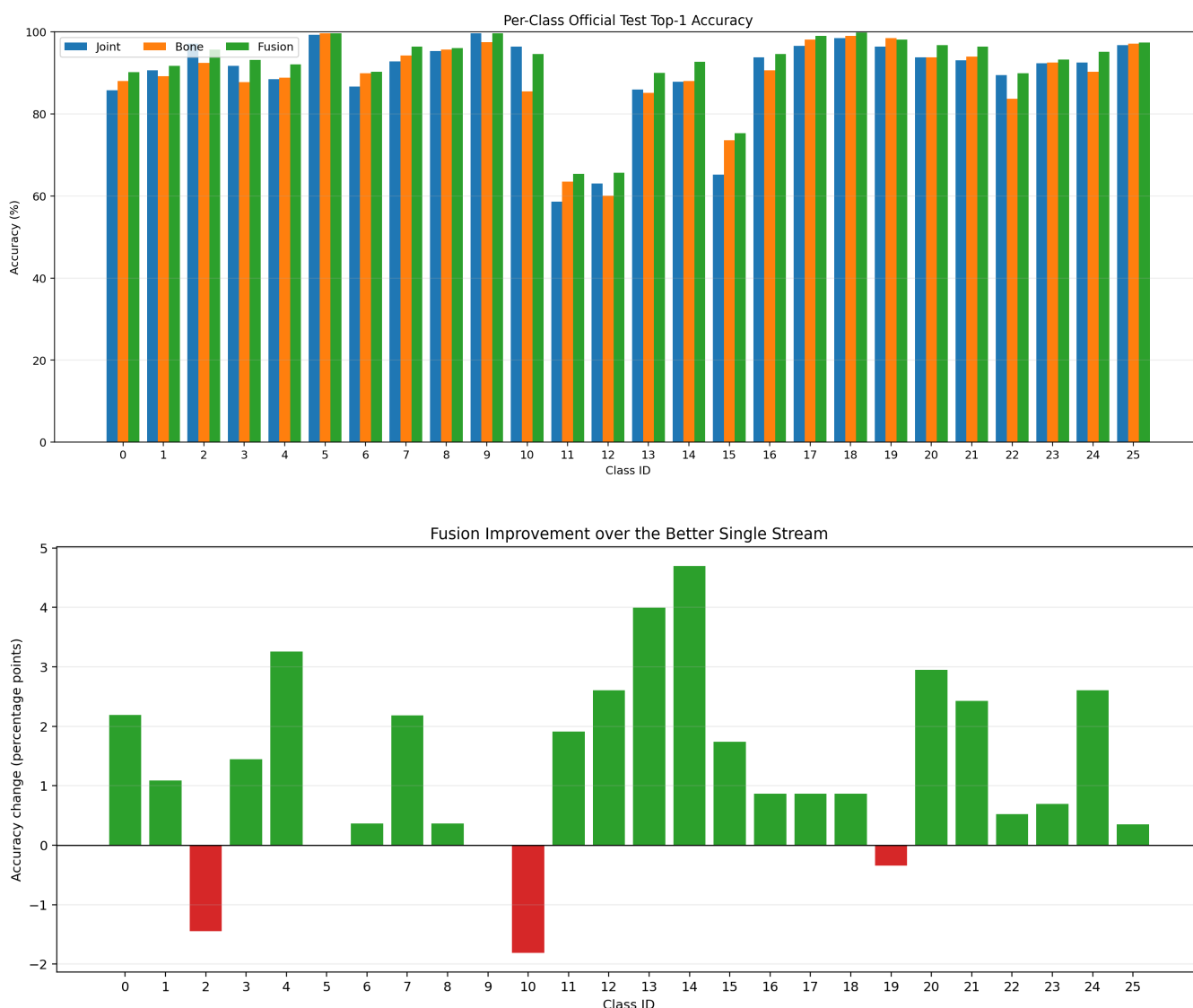
Fusion used the fixed 1:1 sum above. It introduced no additional training, checkpoint, or fusion loss, and no weight was searched on the test set.



Fusion achieved Top-1 91.11% and Top-5 98.82%. This is a **2.62 percentage-point** Top-1 improvement over Joint and **2.79 percentage-point** improvement over Bone.

| Model               | Best Val Top-1 | Best Epoch | Test Top-1 | Test Top-5 | Test Loss |
|---------------------|----------------|------------|------------|------------|-----------|
| Joint               | 94.42%         | 60         | 88.49%     | 98.37%     | 0.400430  |
| Bone                | 93.81%         | 55         | 88.33%     | 98.20%     | 0.430637  |
| Joint + Bone Fusion | N/A            | N/A        | 91.11%     | 98.82%     | N/A       |

## 10. Per-Class Analysis



Relative to the better single stream for each class, the three largest fusion changes were 14 (+4.70 pp), 13 (+3.99 pp), 4 (+3.26 pp). The three smallest changes were 10 (-1.81 pp), 2 (-1.45 pp), 19 (-0.35 pp). Classes with no improvement or a decrease were: 2, 5, 9, 10, 19. Values are empirical class accuracies from the official test labels; class IDs remain 0-25 because no verified action-name mapping was used.

## 11. Main Findings

- Joint: 88.49% test Top-1 and 98.37% Top-5.
- Bone: 88.33% test Top-1 and 98.20% Top-5.
- Fixed 1:1 fusion: 91.11% Top-1 and 98.82% Top-5.

- Fusion improved Top-1 by 2.62 pp over Joint and 2.79 pp over Bone.
- Validation selected Joint epoch 60 and Bone epoch 55.
- Final metrics were independently recomputed from saved scores and ground truth; the test split was not used for selection.

## 12. Limitations

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- Results apply only to Interaction-26 under the NTU120 CSub protocol.
- This is one formal run with fixed seed 1; no multi-seed variance is available.
- Equal 1:1 score fusion is a fixed baseline and need not be the optimal weight.
- Searching fusion weights on the official test set would violate the evaluation protocol and was not performed.

## 13. Reproducibility

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Primary artifacts:

- Saved server configurations: `/mnt/data/6008/CTR-GCN/work_dir/ntu120_interaction26/csub/ctr_gcn_joint_server/config.yaml`, `/mnt/data/6008/CTR-GCN/work_dir/ntu120_interaction26/csub/ctr_gcn_bone_server/config.yaml`
- Main training logs: `/mnt/data/6008/CTR-GCN/work_dir/ntu120_interaction26/csub/ctr_gcn_joint_server/log.txt`, `/mnt/data/6008/CTR-GCN/work_dir/ntu120_interaction26/csub/ctr_gcn_bone_server/log.txt`
- Supplemental console logs for test loss/Top-k: `/mnt/data/6008/CTR-GCN/joint_server_console.log`, `/mnt/data/6008/CTR-GCN/bone_server_console.log`
- Dataset and split: `/mnt/data/6008/data/ntu120/NTU120_CSub_interaction26.npz`, `/mnt/data/6008/data/ntu120/ntu120csubinteraction26_train_val_split.npz`
- Formal work directories: `/mnt/data/6008/CTR-GCN/work_dir/ntu120_interaction26/csub/ctr_gcn_joint_server`, `/mnt/data/6008/CTR-GCN/work_dir/ntu120_interaction26/csub/ctr_gcn_bone_server`
- Best checkpoints: `/mnt/data/6008/CTR-GCN/work_dir/ntu120_interaction26/csub/ctr_gcn_joint_server/best_model.pt`, `/mnt/data/6008/CTR-GCN/work_dir/ntu120_interaction26/csub/ctr_gcn_bone_server/best_model.pt`

- Best-epoch score files: `/mnt/data/6008/CTR-GCN/work_dir/ntu120_interaction26/csub/ctr_gcn_joint_server/epoch60_test_score.pkl`,  
`/mnt/data/6008/CTR-GCN/work_dir/ntu120_interaction26/csub/ctr_gcn_bone_server/epoch55_test_score.pkl`
- Best-epoch per-class CSVs: `/mnt/data/6008/CTR-GCN/work_dir/ntu120_interaction26/csub/ctr_gcn_joint_server/epoch60_test_each_class_acc.csv`,  
`/mnt/data/6008/CTR-GCN/work_dir/ntu120_interaction26/csub/ctr_gcn_bone_server/epoch55_test_each_class_acc.csv`

Rebuild from the project root using:

```
/mnt/data/6008/tools/miniforge3/envs/ctr_gcn_server/bin/python  
reports/ctr_gcn_baseline_v1/generate_report.py
```

The generator writes only inside `reports/ctr_gcn_baseline_v1/`. Its automated checks require complete logs, exact expected key sequences and shapes, finite scores, matching score key sets, 26 classes, expected split sizes, score-vs-log metric agreement, and best-epoch agreement. No runtime-probe or smoke artifact is referenced.